Integral Formula:

1. $\int k\,du = ku + C$
2. $\int u^r du = \dfrac{u^{r+1}}{r+1} + C \quad (r \neq -1)$
3. $\int \dfrac{1}{u} du = \ln|u| + C$
4. $\int e^u du = e^u + C$
5. $\int a^u du = \dfrac{a^u}{\ln a} + C \quad (a > 0,\ a \neq -1)$
6. $\int \sin u\,du = -\cos u + C$
7. $\int \cos u\,du = \sin u + C$
8. $\int \sec^2 u\,du = \tan u + C$
9. $\int \csc^2 u\,du = -\cot u + C$
10. $\int \sec u \tan u\,du = \sec u + C$
11. $\int \csc u \cot u\,du = -\csc u + C$
12. $\int \dfrac{1}{\sqrt{1-u^2}} du = \sin^{-1} u + C$
13. $\int \dfrac{1}{1+u^2} du = \tan^{-1} u + C$
14. $\int \sinh u\,du = \cosh u + C$
15. $\int \cosh u\,du = \sinh u + C$

Integration by Substitution

$$\int f\left[g(x)\right] g'(x)\,dx = \int f(u)\,du$$

$$\int_a^b f\left[g(x)\right] g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

Integration by parts

$$\int u\,dv = uv - \int v\,du$$

Table of Trigonometric substitutions

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin\theta, \quad -\dfrac{\pi}{2} \leqslant \theta \leqslant \dfrac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a \tan\theta, \quad -\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a \sec\theta, \quad 0 \leqslant \theta < \dfrac{\pi}{2}$ or $\pi \leqslant \theta < \dfrac{3\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$